

# Input-Signal Robustness of Multiscale Entropy Biomarkers: The Permutation Entropy Rate as a Uniquely Invariant Complexity Measure for Cardiac Time Series

Martin G. Frasch<sup>1</sup>

Nicolas B. Garnier<sup>2</sup>

February 19, 2026

<sup>1</sup>Department of Obstetrics and Gynecology, University of Washington, Seattle, WA, USA

<sup>2</sup>Laboratoire de Physique, CNRS, ENS de Lyon, Université de Lyon, Lyon, France

## Abstract

Entropy-based complexity measures derived from cardiac interval time series are sensitive to seemingly arbitrary preprocessing choices: whether the input is heart rate (HR) or R-R interval (RRI), and whether the signal is normalized. We systematically evaluate the robustness of three multiscale entropy decay rates—permutation entropy rate ( $h$ ), Sample Entropy (SampEn), and Approximate Entropy (ApEn)—across four input conditions ( $\{\text{HR, RRI}\} \times \{\text{raw, normalized}\}$ ) in 251 segments from 49 children. The  $h$  decay rate shows near-perfect cross-condition agreement (Pearson  $r = 0.983\text{--}0.998$ ) and maintains a significant dose-response relationship ( $p < 0.006$ , Cohen’s  $d > 1.5$ ) in all four conditions. In contrast, SampEn and ApEn decay rates are highly sensitive to input choice ( $r$  as low as  $-0.06$ ), with statistical significance depending on the specific condition tested. We provide a theoretical explanation grounded in the rank-based nature of  $k$ -nearest-neighbor estimation, which is invariant under the monotonic  $\text{HR} \leftrightarrow \text{RRI}$  transform, whereas the absolute tolerance threshold used in template matching is not. These results establish the permutation entropy decay rate as the uniquely robust complexity biomarker for cardiac time series and caution against interpreting template-matching entropy results without explicit robustness verification.

## 1 Introduction

Multiscale entropy analysis has become a standard tool for characterizing the complexity of physiological time series [Costa et al., 2002, 2005]. In cardiac applications, entropy is typically computed from either heart-rate (HR, in beats per minute) or R-R interval (RRI, in milliseconds) time series—two representations of the same underlying process related by the reciprocal transform  $\text{HR} = 60,000/\text{RRI}$ . Additional choices include whether to normalize the signal by its standard deviation, which affects the absolute scale of inter-point distances.

While these preprocessing choices are often treated as inconsequential, they can profoundly affect entropy estimates. Template-matching algorithms (Sample Entropy, Approximate Entropy) use an absolute tolerance threshold  $r_{\text{abs}} = r \times \text{SD}(x)$  that depends on the distributional geometry of the input [Richman and Moorman, 2000]. The nonlinear HR $\leftrightarrow$ RRI transform alters this geometry, potentially changing which templates are counted as matches.

In contrast, k-nearest-neighbor (k-NN) entropy estimators [Kozachenko and Leonenko, 1987, Kraskov et al., 2004] operate on rank-ordered distances, a property that confers invariance under monotonic transforms. This theoretical distinction has practical consequences that, to our knowledge, have not been systematically evaluated in the context of multiscale cardiac complexity analysis.

We address this gap by computing three families of multiscale entropy measures under four input conditions and evaluating their robustness at three levels: (i) segment-level cross-condition correlations, (ii) preservation of dose-response group differences, and (iii) stability of effect sizes. The dataset comprises ambulatory ECG recordings from a cohort study of prenatal steroid exposure [Frasch et al., 2026], providing a clinical context in which biomarker robustness directly affects interpretability.

A companion paper [Garnier and Frasch, 2026] describes the numerical validation of the Python entropy library used for the k-NN computations.

## 2 Methods

### 2.1 Entropy measures

We compute three families of entropy from cardiac interval time series, each capturing different aspects of signal complexity.

#### 2.1.1 Permutation entropy rate ( $h$ )

The permutation entropy rate quantifies the conditional uncertainty of the next value given the past, estimated via k-nearest-neighbor (k-NN) distances using the Kozachenko–Leonenko (KL) estimator [Kozachenko and Leonenko, 1987]:

$$\hat{H}(X) = d \cdot \frac{1}{N_v} \sum_{\substack{i=1 \\ \varepsilon_i > 0}}^N \log(2\varepsilon_i) + \psi(N_v) - \psi(k), \quad (1)$$

where  $\varepsilon_i$  is the  $L_\infty$  (Chebyshev) distance to the  $k$ -th nearest neighbor,  $\psi(\cdot)$  is the digamma function, and  $N_v \leq N$  is the number of points with  $\varepsilon_i > 0$ .

The entropy rate at time scale  $\tau$  is computed via the decomposition [Kraskov et al., 2004]:

$$\hat{h}(\tau) = \hat{H}(X_t) - \hat{I}(X_t; X_{\text{past}}^{(m)}), \quad (2)$$

where  $X_{\text{past}}^{(m)} = (X_{t-\tau}, X_{t-2\tau}, \dots, X_{t-m\tau})$  and  $\hat{I}$  is estimated via the KSG algorithm [Kraskov et al., 2004]. Both  $\hat{H}$  and  $\hat{I}$  are computed on the same unified time-delay embedding to ensure algebraic consistency (see Garnier and Frasch 2026 for implementation details).

### 2.1.2 Sample Entropy (SampEn)

SampEn quantifies signal regularity through template matching [Richman and Moorman, 2000]:

$$\text{SampEn}(m, r) = -\ln\left(\frac{B^{m+1}}{A^m}\right), \quad (3)$$

where  $A^m$  and  $B^{m+1}$  count the number of template pairs matching within tolerance  $r_{\text{abs}} = r \times \text{SD}(x)$  at embedding dimensions  $m$  and  $m + 1$ , respectively, using the  $L_\infty$  norm and excluding self-matches.

### 2.1.3 Approximate Entropy (ApEn)

ApEn follows the same template-matching framework but includes self-matches in the counting [Pincus, 1991], making it less sensitive to short data but introducing a bias toward lower values.

## 2.2 Multiscale analysis

### 2.2.1 Scale sweep

Entropy is computed at time scales ranging from 0.5 to 5.0 s in steps of  $d\tau = 0.04$  s (i.e., `dstride` = 10 samples at 250 Hz), yielding approximately 113 scale points per segment.

For SampEn and ApEn, multiscale analysis uses non-overlapping block averaging (coarse-graining; Costa et al. 2002):

$$y_j^{(\tau)} = \frac{1}{\tau} \sum_{i=(j-1)\tau+1}^{j\tau} x_i. \quad (4)$$

For the entropy rate  $h$ , multiscale behavior is captured implicitly through the stride parameter  $\tau$  in the time-delay embedding (Eq. 2), operating on the original signal with progressively wider lags rather than on a downsampled signal.

### 2.2.2 Summary statistics

From each entropy-versus-scale curve, four summary statistics are extracted:

- **AUC**: area under the curve (trapezoidal integration)
- **max**: maximum entropy value across scales
- **max\_t**: scale at which the maximum occurs
- **dec\_rate**: rate of decrease from the maximum to the final scale

This yields 12 features per segment:  $\{h, \text{SampEn}, \text{ApEn}\} \times \{\text{AUC}, \text{max}, \text{max\_t}, \text{dec\_rate}\}$ .

## 2.3 Computation parameters

### 2.4 Input conditions

All entropy measures were computed under four input-signal variants (Table 2), each with and without temporal smoothing (8 total configurations).

Table 1: Entropy computation parameters.

| Parameter                   | Value                  | Notes                                         |
|-----------------------------|------------------------|-----------------------------------------------|
| Embedding dimension $m$     | 2                      | Entropy rate and SampEn/ApEn                  |
| k-NN neighbors $k$          | 5                      | KL estimator                                  |
| Tolerance $r$ (SampEn/ApEn) | $0.2 \times \text{SD}$ | Fixed across scales                           |
| Distance metric             | $L_\infty$ (Chebyshev) | Both k-NN and template matching               |
| Scale range                 | 0.5–5.0 s              | $d\tau = 0.04$ s ( <code>dstride</code> = 10) |
| $N_{\text{eff}}$            | 4096                   | Points per realization                        |
| $N_{\text{real}}$           | 3                      | Realizations averaged                         |

Table 2: Four input conditions for robustness analysis.

| Label                      | Signal             | Normalization    | Transform from RRI |
|----------------------------|--------------------|------------------|--------------------|
| $\text{HR}_{\text{raw}}$   | Heart rate (bpm)   | None             | 60,000/RRI         |
| $\text{HR}_{\text{norm}}$  | Heart rate         | $\div \text{SD}$ | Monotonic          |
| $\text{RRI}_{\text{raw}}$  | R-R intervals (ms) | None             | Identity           |
| $\text{RRI}_{\text{norm}}$ | R-R intervals      | $\div \text{SD}$ | Monotonic          |

## 2.5 Dataset

We analyze 251 five-minute ECG segments (250 Hz) from 49 children aged 8 years, drawn from a prospective cohort study of prenatal maternal steroid exposure [Frasch et al., 2026]. Subjects are classified into three groups based on cumulative maternal steroid dose: low dose ( $< 5$  g,  $N = 12\text{--}15$ ), high dose ( $\geq 5$  g,  $N = 13\text{--}15$ ), and controls ( $N = 24\text{--}30$ ), with exact counts varying by segment due to quality exclusions. R-R intervals are extracted via ensemble R-peak detection with signal quality index (SQI) filtering.

## 2.6 Statistical analysis

### 2.6.1 Cross-condition correlations

Segment-level Pearson correlations between decay rates computed under different input conditions.

### 2.6.2 Dose-response preservation

Kruskal-Wallis tests comparing three dose groups on the mental arithmetic segment (M3\_bis\_M4, peak cognitive load), repeated under each input condition.

### 2.6.3 Pairwise effect sizes

Mann-Whitney  $U$  tests with Cohen’s  $d$  for the low-dose vs. high-dose comparison under each condition.

### 3 Results

#### 3.1 Cross-condition correlations

Table 3 shows segment-level Pearson correlations ( $N = 251$ ) between decay rates computed under the four input conditions.

Table 3: Cross-condition Pearson correlations for entropy decay rates. The  $h$  decay rate shows near-perfect agreement ( $r > 0.98$ ) across all conditions. SampEn and ApEn decay rates are highly sensitive to input choice.

| Pair                                      | $h$ decay rate     | SampEn decay rate | ApEn decay rate |
|-------------------------------------------|--------------------|-------------------|-----------------|
|                                           | $r$                | $r$               | $r$             |
| HR <sub>raw</sub> vs HR <sub>norm</sub>   | <b>0.997</b>       | 0.713             | 0.091           |
| HR <sub>raw</sub> vs RRI <sub>raw</sub>   | <b>0.983</b>       | 0.502             | 0.099           |
| HR <sub>raw</sub> vs RRI <sub>norm</sub>  | <b>0.985</b>       | 0.698             | 0.089           |
| HR <sub>norm</sub> vs RRI <sub>raw</sub>  | <b>0.985</b>       | 0.316             | −0.045          |
| HR <sub>norm</sub> vs RRI <sub>norm</sub> | <b>0.986</b>       | 0.953             | 0.980           |
| RRI <sub>raw</sub> vs RRI <sub>norm</sub> | <b>0.998</b>       | 0.310             | −0.060          |
| Range                                     | <b>0.983–0.998</b> | 0.310–0.953       | −0.060–0.980    |

The  $h$  decay rate is virtually invariant to input choice, with all pairwise correlations exceeding 0.98. In contrast, SampEn decay rate correlations range from 0.31 to 0.95, and ApEn decay rate correlations range from −0.06 to 0.98. For both template-matching measures, only the HR<sub>norm</sub> vs RRI<sub>norm</sub> pair achieves reasonable agreement ( $r > 0.95$ ), while comparisons involving raw signals show moderate to negligible correlations.

#### 3.2 Dose-response preservation

Table 4 presents Kruskal-Wallis tests comparing three dose groups on the mental arithmetic segment under each input condition.

Table 4: Dose-response analysis (Kruskal-Wallis) on the M3\_bis\_M4 segment across input conditions. Only the  $h$  decay rate reaches significance ( $p < 0.01$ ) in all four conditions.

| Condition           | $h$ decay rate |              | SampEn decay rate |              | ApEn decay rate |       |
|---------------------|----------------|--------------|-------------------|--------------|-----------------|-------|
|                     | $H$            | $p$          | $H$               | $p$          | $H$             | $p$   |
| HR <sub>raw</sub>   | 10.16          | <b>0.006</b> | 12.49             | <b>0.002</b> | 1.90            | 0.387 |
| HR <sub>norm</sub>  | 10.59          | <b>0.005</b> | 4.09              | 0.129        | 4.41            | 0.110 |
| RRI <sub>raw</sub>  | 10.18          | <b>0.006</b> | 4.51              | 0.105        | 0.37            | 0.832 |
| RRI <sub>norm</sub> | 10.69          | <b>0.005</b> | 4.49              | 0.106        | 4.51            | 0.105 |

The  $h$  decay rate yields consistent  $H$  statistics (10.16–10.69) and significant  $p$  values ( $p < 0.006$ ) across all four conditions. The SampEn decay rate is significant only under HR<sub>raw</sub> ( $p = 0.002$ ) and fails to reach significance in all other conditions. The ApEn decay rate is not significant in any condition.

### 3.3 Pairwise comparisons

Table 5 shows Mann-Whitney  $U$  tests for the low-dose vs. high-dose comparison.

Table 5: Low-dose vs. high-dose pairwise comparisons (Mann-Whitney  $U$ ) on the M3\_bis\_M4 segment. The  $h$  decay rate shows consistently large effects ( $d > 1.5$ ,  $p < 0.005$ ) in all conditions.

| Condition           | $h$ decay rate |              |      | SampEn decay rate |                  |      |
|---------------------|----------------|--------------|------|-------------------|------------------|------|
|                     | $U$            | $p$          | $d$  | $U$               | $p$              | $d$  |
| HR <sub>raw</sub>   | 25             | <b>0.004</b> | 1.53 | 14                | <b>&lt;0.001</b> | 1.97 |
| HR <sub>norm</sub>  | 23             | <b>0.003</b> | 1.57 | 42                | 0.054            | 0.93 |
| RRI <sub>raw</sub>  | 25             | <b>0.004</b> | 1.57 | 21                | 0.062            | 0.97 |
| RRI <sub>norm</sub> | 24             | <b>0.004</b> | 1.66 | 41                | 0.047            | 0.98 |

The  $h$  decay rate shows large, stable effects (Cohen’s  $d = 1.53$ – $1.66$ ) across all conditions, with  $p < 0.005$  uniformly. The SampEn decay rate achieves the largest single effect size ( $d = 1.97$ ) under HR<sub>raw</sub>, but this effect does not generalize: in the remaining three conditions, the  $p$  values hover near 0.05 and the effect sizes drop to  $d \approx 0.95$ .

### 3.4 Summary of robustness properties

Table 6: Summary of robustness properties across entropy decay rates.

| Property                                      | $h$ dec. rate    | SampEn dec. rate | ApEn dec. rate |
|-----------------------------------------------|------------------|------------------|----------------|
| Cross-condition $r$ (min)                     | <b>0.983</b>     | 0.310            | −0.060         |
| Significant in all 4 (KW)                     | <b>Yes (4/4)</b> | No (1/4)         | No (0/4)       |
| Signif. low vs. high, all 4 (MW)              | <b>Yes (4/4)</b> | No (1–2/4)       | No (0–1/4)     |
| Effect size range ( $d$ )                     | 1.53–1.66        | 0.93–1.97        | 0.05–0.93      |
| HR <sub>norm</sub> vs RRI <sub>norm</sub> $r$ | <b>0.986</b>     | 0.953            | 0.980          |

## 4 Discussion

### 4.1 Theoretical basis: k-NN rank invariance

The robustness of the  $h$  decay rate follows from the rank-based nature of the k-NN estimator. The KL estimator (Eq. 1) depends on neighbor distances  $\varepsilon_i$  through their logarithms. Under a monotonic transform  $g(x)$ :

1. The *rank ordering* of inter-point distances is preserved—the same points remain nearest neighbors.
2. The distances scale by the local derivative  $|g'|$ , adding a term  $\overline{\ln |g'|}$  to the entropy estimate.
3. Entropy *differences* across scales—which determine the decay rate—are *invariant*, because the constant offset cancels in the subtraction.

The HR $\leftrightarrow$ RRI conversion ( $g(x) = 60,000/x$ ) is a strictly monotonic (decreasing) transform. Normalization by standard deviation ( $x \mapsto x/\sigma$ ) is also monotonic. Therefore, while absolute entropy values differ across conditions (by a constant  $\ln |g'|$ ), the *shape* of the entropy-versus-scale curve—and hence its decay rate—is preserved. This explains the cross-condition correlations  $r > 0.98$  observed for the  $h$  decay rate.

## 4.2 Template-matching sensitivity

SampEn and ApEn use an absolute tolerance threshold  $r_{\text{abs}} = 0.2 \times \text{SD}(x)$ . Under the nonlinear HR $\leftrightarrow$ RRI transform:

1. The standard deviation changes nonlinearly:  $\text{SD}(60,000/x) \neq 60,000 \cdot \text{SD}(x)/\bar{x}^2$  in general.
2. The tolerance  $r_{\text{abs}}$  captures a *different proportion* of the signal’s distributional geometry.
3. The set of matching templates changes, sometimes dramatically.

This effect is asymmetric across the HR/RRI range: at low HR values (long RRI), a given RRI tolerance corresponds to a small HR tolerance, while at high HR values the same RRI tolerance corresponds to a large HR tolerance. The low cross-condition correlations for SampEn ( $r$  as low as 0.31) and ApEn ( $r$  as low as  $-0.06$ ) directly reflect this sensitivity.

## 4.3 Normalization as partial remedy

Normalization by standard deviation partially mitigates the template-matching sensitivity. The HR $_{\text{norm}}$  vs RRI $_{\text{norm}}$  correlations are high for all three measures ( $r = 0.953$ – $0.986$ ; Table 3, last row). However, this agreement only holds between normalized variants; comparisons involving raw signals remain unstable. Moreover, normalization does not restore the dose-response relationship: even under RRI $_{\text{norm}}$ , only the  $h$  decay rate retains significance (Table 4).

This observation has important practical implications: while normalization improves reproducibility of template-matching measures, it does not make them robust biomarkers. The  $h$  decay rate, by contrast, yields the same statistical conclusion regardless of preprocessing choice.

## 4.4 The SampEn “HR $_{\text{raw}}$ anomaly”

The SampEn decay rate achieves its strongest result ( $H = 12.49$ ,  $p = 0.002$ ,  $d = 1.97$ ) specifically under HR $_{\text{raw}}$ —exceeding even the  $h$  decay rate in that single condition. This isolated finding does not replicate under any other input condition. We interpret it as an artifact of the particular distributional geometry of raw heart rate: the reciprocal transform compresses long RRI values (low HR) and expands short RRI values (high HR), creating a specific template-matching landscape that happens to amplify group differences. Had only HR $_{\text{raw}}$  been tested, SampEn decay rate would have appeared to be the stronger biomarker—a cautionary example of condition-dependent findings.

## 4.5 Relation to implementation validation

A companion paper [Garnier and Frasch, 2026] validates the Python implementation of the k-NN entropy estimators against the reference C/ANN library, identifying and correcting three implementation-level discrepancies (zero-distance handling, distance-metric mismatch, embedding alignment). The present work complements that validation by demonstrating that k-NN entropy measures are not only numerically correct but also *methodologically robust*—their biological conclusions are invariant to preprocessing choices that fundamentally alter template-matching results.

The two papers share a common insight: the rank-based nature of k-NN estimation is the key property. In Garnier and Frasch [2026], it ensures that the corrected implementation produces identical neighbor rankings as the C reference. Here, it ensures that the same neighbor rankings—and hence the same entropy decay rates—emerge regardless of whether the input is HR or RRI, raw or normalized.

## 4.6 Limitations

This analysis evaluates robustness across input-signal transformations but does not address other sources of variability such as segment length, R-peak detection method, or noise contamination. The dose-response analysis is limited to a single cognitive-stress segment (mental arithmetic) in a modest cohort ( $N = 49$ ), and replication in larger datasets is needed. The theoretical rank-invariance argument is exact only for strictly monotonic transforms; nonlinear transforms that change the rank order (e.g., absolute value) would break the invariance.

## 5 Conclusion

Among three families of multiscale entropy decay rates, only the permutation entropy rate ( $h$  decay rate) is robust to the choice of input signal representation. Its cross-condition correlations exceed  $r = 0.98$ , and it maintains a large, significant dose-response effect ( $d > 1.5$ ,  $p < 0.005$ ) under all four tested conditions. The theoretical basis is the rank invariance of k-NN distance estimation under monotonic transforms—a property absent from the template-matching framework used by SampEn and ApEn.

We recommend that studies reporting entropy-based cardiac complexity measures:

1. Use k-NN-based entropy rates as the primary complexity biomarker.
2. Verify robustness across input conditions before interpreting template-matching results.
3. Report the specific input condition (HR vs. RRI, raw vs. normalized) to enable reproducibility.

All analysis code and data are available at <https://github.com/martinfrasch/entropy> and [https://github.com/martinfrasch/ECG\\_steroids](https://github.com/martinfrasch/ECG_steroids).

## Acknowledgments

The robustness analysis and manuscript preparation were performed with assistance from Claude Code (Anthropic).



## References

- M. Costa, A. L. Goldberger, and C.-K. Peng. Multiscale entropy analysis of complex physiologic time series. *Physical Review Letters*, 89(6):068102, 2002. doi: 10.1103/PhysRevLett.89.068102.
- M. Costa, A. L. Goldberger, and C.-K. Peng. Multiscale entropy analysis of biological signals. *Physical Review E*, 71(2):021906, 2005. doi: 10.1103/PhysRevE.71.021906.
- M. G. Frasch et al. Prenatal steroid exposure and cardiac complexity in children. *In preparation*, 2026. Main study manuscript.
- N. B. Garnier and M. G. Frasch. Numerical validation of a pure Python implementation of k-NN entropy estimation against a reference C/ANN library. *In preparation*, 2026. Companion paper on implementation validation.
- L. F. Kozachenko and N. N. Leonenko. Sample estimate of the entropy of a random vector. *Problemy Peredachi Informatsii*, 23(2):9–16, 1987.
- A. Kraskov, H. Stögbauer, and P. Grassberger. Estimating mutual information. *Physical Review E*, 69(6):066138, 2004. doi: 10.1103/PhysRevE.69.066138.
- S. M. Pincus. Approximate entropy as a measure of system complexity. *Proceedings of the National Academy of Sciences*, 88(6):2297–2301, 1991. doi: 10.1073/pnas.88.6.2297.
- J. S. Richman and J. R. Moorman. Physiological time-series analysis using approximate entropy and sample entropy. *American Journal of Physiology—Heart and Circulatory Physiology*, 278(6):H2039–H2049, 2000. doi: 10.1152/ajpheart.2000.278.6.H2039.